Task 12 - Cellular automata - forest fire algorithm

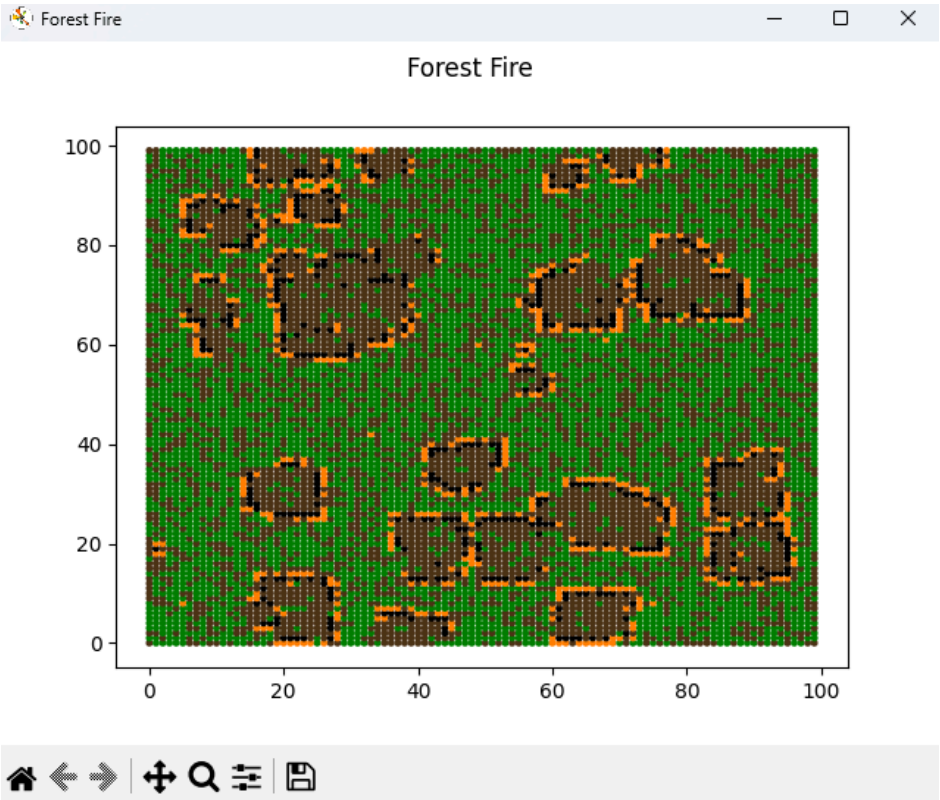
Forest fire model is a special use case of Cellular Automaton algorithm which is represented in second dimension by following rules:

1. An empty area or a burnt tree is replaced with probability *p* either by a live one or remains empty
2. A tree will catch fire if its neighbor is on fire
3. If no tree in the neighbor is burning, then it will start burning with probability *f*
4. If the tree burns, it will eventually burn out and stay burned

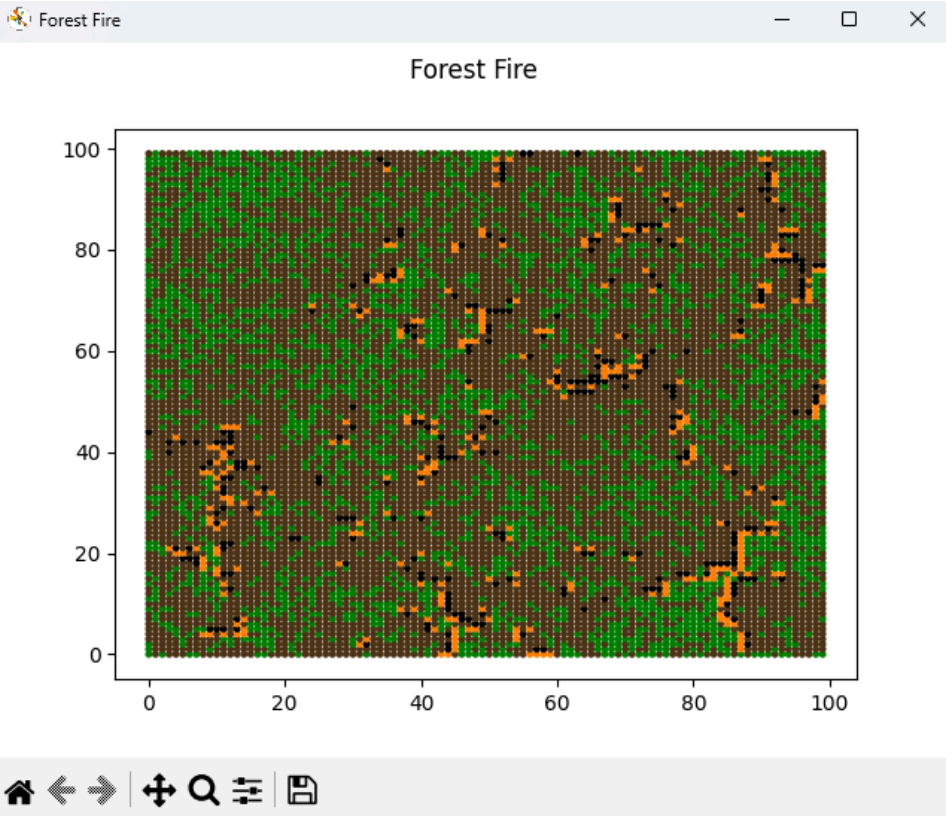
As an example of values you can use *p = 0.05; f = 0.001* and forest *density = 0.5*. For a neighborhood the von Neumann is used, if you want you can try Moore as well.

Instructions

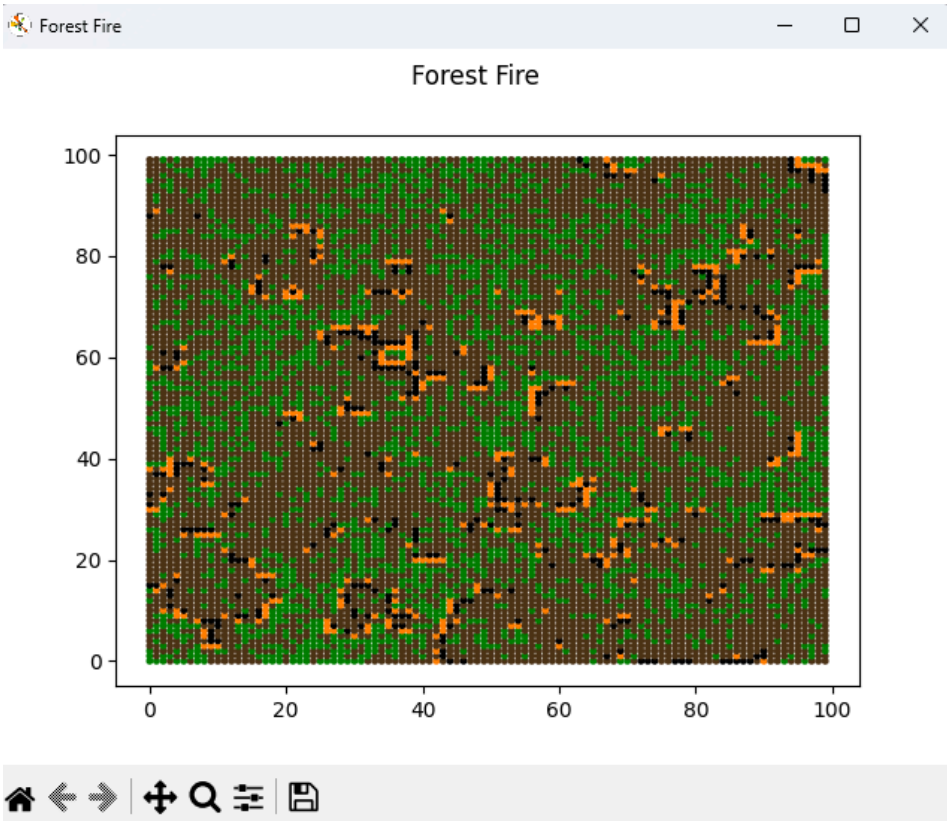
Implement the algorithm in a way that it will work infinitely in the animation. You can use matplotlib or any other visualization package to do this.



Example 1



Example 2



Example 3